

Name: _____





BOOTSTRAP

Equity • Scale • Rigor

Brought to you by the Bootstrap team:

- Emmanuel Schanzer
- Kathi Fiser
- Shriram Krishnamurthi
- Dorai Sitaram
- Joe Politz
- Ben Lerner
- Nancy Pfenning
- Flannery Denny
- Rachel Tabak

Computing Decision Trees

```
# build-tree :: ( Table , List<String> , String , Number ) -> DecisionTree
                  Training Data      Columns to Split On      "Answer key" Column      Maximum Depth
```

Return to your copy of the [Decision Tree Starter File](#).

Building One-Level Decision Trees from Categorical and Quantitative Splits

In the Definitions Area, define **tree-c1** and to be a 1-level tree using only the **categorical** "tail" column, and click "Run":

```
tree-c1 = build-tree(training, [list: "tail"], "species", 1)
```

1) Evaluate **tree-c1**. If an animals has no tail, it will predict: _____

2) Evaluate **dt-code(tree-c1)**. Where have we seen this code before? _____

In the Definitions Area, define **tree-q1** to be a 1-level tree using only the **quantitative** "pounds" column, and click "Run":

```
tree-q1 = build-tree(training, [list: "pounds"], "species", 1)
```

3) Evaluate **tree-q1**. What split did it come up with? _____

In the Definitions Area, define **tree-any** to be a 1-level tree using the **any** column, and click "Run":

```
tree-any1 = build-tree(training, [list: "sex", "tail", "mammal", "swims", "pounds"], "species", 1)
```

4) Which column would **you** choose for the **root node**? _____. Evaluate **tree-any1**. Which one did Pyret choose? _____

Building Multi-level Decision Trees from Categorical and Quantitative Splits

Add the following definitions to your Definitions Area and click "Run":

```
tree-c2 = build-tree(training, [list: "tail"], "species", 2)
tree-q2 = build-tree(training, [list: "pounds"], "species", 2)
tree-any2 = build-tree(training, [list: "sex", "tail", "mammal", "swims", "pounds"], "species", 2)
```

5) In the code above, we allowed **tree-c2** to have a maximum depth of _____. The resulting tree actually had _____ level. Why?

Add the following definitions to your Definitions Area and click "Run":

```
tree-q3 = build-tree(training, [list: "pounds"], "species", 3)
tree-q4 = build-tree(training, [list: "pounds"], "species", 4)
tree-q5 = build-tree(training, [list: "pounds"], "species", 5)
tree-any3 = build-tree(training, [list: "sex", "tail", "mammal", "swims", "pounds"], "species", 3)
tree-any4 = build-tree(training, [list: "sex", "tail", "mammal", "swims", "pounds"], "species", 4)
tree-any5 = build-tree(training, [list: "sex", "tail", "mammal", "swims", "pounds"], "species", 5)
```

6) We've produced three of the trees below. How are they similar? How are they different? _____

tree-q1	tree-q2	tree-q3
<pre>pounds < 9.55? ├true→ cat └false→ dog</pre>	<pre>pounds < 9.55? ├true pounds < 0.19? │├true→ snail │└false→ cat └false→ dog</pre>	<pre>pounds < 9.55? ├true pounds < 0.19? │├true→ snail │└false pounds < 1.95? │├true→ tarantula │└false→ cat └false→ dog</pre>

Be sure to save your copy of the [Decision Tree Starter File](#). Then turn to [Comparing Decision Trees](#).

Comparing Decision Trees

To complete this page, you will need to refer to [Computing Decision Trees](#) and your saved copy of the [Decision Tree Starter File](#).

Comparing Trees for our Training Dataset

tree	max levels	levels	# correct for training	# correct for testing
tree-c1	1	1		
tree-q1	1	1		
tree-any1	1	1		
tree-c2	2			
tree-q2	2			
tree-any2	2			
tree-q3	3			
tree-any3	3			
tree-q4	4			
tree-any4	4			
tree-q5	5			
tree-any5	5			
tree-q6	6			

- Evaluate each of the decision trees we've defined and record how many levels deep it is in the **levels** column.
- **Classify** each of the Decisions trees using the **training** dataset, e.g. `classify(training, "species", tree-c1)`, and complete the **number correct for training dataset** column.

1) What do you notice? What do you wonder? _____

2) Did allowing for more levels in the tree always make the tree better? _____

3) If we were using one column, which would do a better job of building a tree for the training data: categorical or quantitative? _____

Using our Classifiers with a Testing Dataset

We built our classifiers using the **training** dataset. Let's see how well they do predicting **"species"** for a new dataset called **testing**.

4) **Classify** each of the Decisions trees using the **testing** dataset, e.g. `classify(testing, "species", tree-c1)`, and complete the **number correct for training dataset** column.

5) Which two models did the besst job of predicting species for our **training** dataset? _____ and _____

6) How did their success rates compare with our **testing** dataset? _____

7) Look carefully at the number correct for our **tree-q** decision trees built on the pounds column as the number of allowable levels increases. What do you notice? _____

Confusion Matrices

```
fun swims-classifier(r):  
  if r["swims"] == true:  
    "dog"  
  else:  
    "cat"  
end
```

- Open your copy of the [Decision Tree Starter File](#), add the definition for swims-classifier to your Definitions Area and click "Run".
- 1) Use the classify functions with the swims-classifier to complete the confusion matrix table below.
- Fill in the first blank ("out of X")
 - Then, convert the count of correct predictions into a percentage
 - Finally, convert the percentage of correct predictions into a decimal

Computing the Report by Hand

actual-species	predicted-cat	predicted-dog	predicted-lizard	predicted-snail	predicted-tarantula
"cat"	$\frac{4}{\text{out of } 5} = \frac{80}{\% \text{ pct}} = \frac{0.8}{\text{decimal}}$	$\frac{\text{out of } 5}{\text{out of } 5} = \frac{\% \text{ pct}}{\% \text{ pct}} = \frac{\text{decimal}}{\text{decimal}}$	$\frac{\text{out of } 5}{\text{out of } 5} = \frac{\% \text{ pct}}{\% \text{ pct}} = \frac{\text{decimal}}{\text{decimal}}$	$\frac{\text{out of } 5}{\text{out of } 5} = \frac{\% \text{ pct}}{\% \text{ pct}} = \frac{\text{decimal}}{\text{decimal}}$	$\frac{\text{out of } 5}{\text{out of } 5} = \frac{\% \text{ pct}}{\% \text{ pct}} = \frac{\text{decimal}}{\text{decimal}}$
"dog"	$\frac{\text{out of } 4}{\text{out of } 4} = \frac{\% \text{ pct}}{\% \text{ pct}} = \frac{\text{decimal}}{\text{decimal}}$	$\frac{\text{out of } 4}{\text{out of } 4} = \frac{\% \text{ pct}}{\% \text{ pct}} = \frac{\text{decimal}}{\text{decimal}}$	$\frac{\text{out of } 4}{\text{out of } 4} = \frac{\% \text{ pct}}{\% \text{ pct}} = \frac{\text{decimal}}{\text{decimal}}$	$\frac{\text{out of } 4}{\text{out of } 4} = \frac{\% \text{ pct}}{\% \text{ pct}} = \frac{\text{decimal}}{\text{decimal}}$	$\frac{\text{out of } 4}{\text{out of } 4} = \frac{\% \text{ pct}}{\% \text{ pct}} = \frac{\text{decimal}}{\text{decimal}}$
"lizard"	$\frac{\text{out of } 2}{\text{out of } 2} = \frac{\% \text{ pct}}{\% \text{ pct}} = \frac{\text{decimal}}{\text{decimal}}$	$\frac{\text{out of } 2}{\text{out of } 2} = \frac{\% \text{ pct}}{\% \text{ pct}} = \frac{\text{decimal}}{\text{decimal}}$	$\frac{\text{out of } 2}{\text{out of } 2} = \frac{\% \text{ pct}}{\% \text{ pct}} = \frac{\text{decimal}}{\text{decimal}}$	$\frac{\text{out of } 2}{\text{out of } 2} = \frac{\% \text{ pct}}{\% \text{ pct}} = \frac{\text{decimal}}{\text{decimal}}$	$\frac{\text{out of } 2}{\text{out of } 2} = \frac{\% \text{ pct}}{\% \text{ pct}} = \frac{\text{decimal}}{\text{decimal}}$
"snail"	$\frac{\text{out of } 2}{\text{out of } 2} = \frac{\% \text{ pct}}{\% \text{ pct}} = \frac{\text{decimal}}{\text{decimal}}$	$\frac{\text{out of } 2}{\text{out of } 2} = \frac{\% \text{ pct}}{\% \text{ pct}} = \frac{\text{decimal}}{\text{decimal}}$	$\frac{\text{out of } 2}{\text{out of } 2} = \frac{\% \text{ pct}}{\% \text{ pct}} = \frac{\text{decimal}}{\text{decimal}}$	$\frac{\text{out of } 2}{\text{out of } 2} = \frac{\% \text{ pct}}{\% \text{ pct}} = \frac{\text{decimal}}{\text{decimal}}$	$\frac{\text{out of } 2}{\text{out of } 2} = \frac{\% \text{ pct}}{\% \text{ pct}} = \frac{\text{decimal}}{\text{decimal}}$
"tarantula"	$\frac{\text{out of } 2}{\text{out of } 2} = \frac{\% \text{ pct}}{\% \text{ pct}} = \frac{\text{decimal}}{\text{decimal}}$	$\frac{\text{out of } 2}{\text{out of } 2} = \frac{\% \text{ pct}}{\% \text{ pct}} = \frac{\text{decimal}}{\text{decimal}}$	$\frac{\text{out of } 2}{\text{out of } 2} = \frac{\% \text{ pct}}{\% \text{ pct}} = \frac{\text{decimal}}{\text{decimal}}$	$\frac{\text{out of } 2}{\text{out of } 2} = \frac{\% \text{ pct}}{\% \text{ pct}} = \frac{\text{decimal}}{\text{decimal}}$	$\frac{\text{out of } 2}{\text{out of } 2} = \frac{\% \text{ pct}}{\% \text{ pct}} = \frac{\text{decimal}}{\text{decimal}}$

Let Pyret Make the Report

2) Have Pyret generate a confusion-matrix for our swims-classifier using the testing table. What can we learn? How does the swim-classifier performance with the testing table compare to its performance with the training table? _____

These materials were developed partly through support of the National Science Foundation (awards 1042210, 1535276, 1648684, and 1738598) and are licensed under a Creative Commons 4.0 Unported License. Based on a work at www.BootstrapWorld.org. Permissions beyond the scope of this license may be available by contacting contact@BootstrapWorld.org.